📘 MODULE 6: SPL FROM ZERO

6.1 Concept

SPL (Search Processing Language) is Splunk's search language. A search is a **pipeline**: the first part retrieves events, and each | passes the results to the next command.

index=main error | stats count by host | sort - count

└ get events ┘ └ summarize ┘ └ order ┘

6.2 Why it matters

Every SOC task (hunting failed logins, finding top talkers, building a dashboard) is an SPL pipeline.

6.3 The pipe, explained

- **Before the first pipe:** the base search. Splunk reads events from disk. Be specific here, because this is where performance is won or lost.
- **After each pipe:** a command that takes the previous output as input and produces new output.
- The output can change shape: events become a table of statistics after stats, and later commands can only see what earlier commands kept.

6.4 Search basics

Search	Meaning
index=*	Every index you can access (excludes _ internal ones)
index=main	All events in main
index=main error	Events in main containing the word *error* anywhere in _raw
index=main host=myhost	Events from that host
index=main "Failed password"	Exact phrase (quotes needed for spaces)
index=main error OR fail*	Boolean OR, plus wildcard *
index=main error NOT debug	Exclude events containing *debug*
index=main (error OR warn) host=web1	Parentheses group logic

Rules to remember

- Words next to each other mean **AND** implicitly.
- Operators AND, OR, NOT must be **UPPERCASE**.
- Field **names** are case-sensitive (host, not Host). Search **values** are case-insensitive (error matches ERROR).
- field!=value excludes events where the field is missing. NOT field=value keeps them. This difference causes real bugs.
- A leading wildcard (*error) is slow. Avoid it.

Comments use triple backticks:

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index=main ```this is a comment``` error

6.5 Core commands

Query

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index=_internal sourcetype=splunkd_access

| table _time clientip user method status uri_path

What it does: shows selected fields as a clean table.
Line-by-line: the first line retrieves events. table keeps only the listed fields, in that order, as columns.
Expected result: a table with those columns.
Common mistakes: commas are optional but spaces are required. Fields that don't exist show blank columns.
Interview explanation: "table is for presentation. It's a transforming-style command, so I use it last."

Command reference

Command	Purpose	Example
table	Show chosen fields as columns	| table _time host status
fields	Keep (+) or remove (-) fields. **Faster than table mid-search** because it drops unneeded fields early	| fields host, status / | fields - _raw

Command	Purpose	Example
rename	Change a field's display name	| rename clientip AS "Source IP"
where	Filter with expressions (case-sensitive, supports comparisons and functions)	| where status>=400
search	Filter later in the pipeline, same syntax as the base search	| search status=404
sort	Order results. - means descending	| sort - count or | sort 0 - count
head	First N results (default 10)	| head 5
tail	Last N results	| tail 5
dedup	Keep the first event per unique value	| dedup clientip
top	Most common values, with count and percent	| top limit=5 clientip
rare	Least common values	| rare limit=5 uri_path
stats	Aggregate (Module 7)	| stats count by host
chart	Aggregate for charting: one field on each axis	| chart count over host by status
timechart	Aggregate over time (Module 9)	| timechart span=1h count

Key distinctions

- **search vs where:** search is simple field/keyword matching. where evaluates expressions (where count>10, where like(user,"adm%")). In where, strings need double quotes, and comparisons against another field use the bare field name.
- **sort limit:** sort returns only **10,000 results by default**. Use sort 0 - count to remove the limit.
- **top vs stats count ... | sort:** top is a shortcut that adds a percent column and defaults to 10 rows.
- **dedup keeps the first event it sees**, which is usually the newest, because events are returned newest first.

Query: top 10 sources of requests

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index=_internal sourcetype=splunkd_access

| top limit=10 clientip

What it does: counts events per clientip and shows the 10 most frequent, with percent.
Expected result: columns clientip, count, percent.
Common mistake: using top on a field with a huge number of unique values, which is slow.
Interview explanation: "top is a quick frequency analysis. For more control I use stats count by followed by sort."

6.6 Hands-on

1. List the 20 newest events in _internal that contain ERROR.
2. Show only _time, host, component, log_level for those events (sourcetype=splunkd).
3. Find the 5 least common component values (rare).
4. Run a search with a comment in it.

6.7 Troubleshooting

Problem	Cause
table columns empty	Field name typo, or the field doesn't exist in that sourcetype
or not working	Must be uppercase OR
sort seems to miss results	Default 10,000 cap, so use sort 0
Search is very slow	index=* over All time, so narrow the index and time

6.8 Interview Q&A

- **What does the pipe do?** It passes the output of the left command as input to the right command.
- **search vs where?** search is keyword/field matching, where evaluates expressions and is case-sensitive.
- **fields vs table?** fields removes fields early and speeds up the search. table formats the output.
- **top vs rare?** Most vs least frequent values.

6.9 Revision

Pipe = output becomes input · filter in the base search first · OR/AND/NOT uppercase · sort 0 for unlimited · fields early, table last.